

# BusMind: An Emotion-Aware AI Study Companion for Bus Commuters

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## ABSTRACT:

BusMind is a mobile AI study companion designed for students who spend a significant amount of time travelling by bus. Instead of treating commuting as unused time, the system attempts to convert short and fragmented travel intervals into manageable learning sessions while also considering the learner's current context. The application integrates facial-expression-based emotion detection, mood-aware music recommendation, smart flashcards, voice-based note summarization, posture monitoring, learning analytics, and graph-based study-path organization within a Flutter application. The system uses a Vision Transformer-based facial-expression model to obtain a broad expression category, which is subsequently converted into an application-level mood state. This mood information is combined with user preferences and session context before generating recommendations. The learning component uses locally stored flashcards and a spaced-review mechanism to prioritize cards that require attention. Voice input is converted into text using device-supported speech recognition and can subsequently be summarized into concise study material. Accelerometer and gyroscope readings are processed together to identify persistent device-orientation patterns and provide posture-awareness reminders. The application follows a modular architecture in which authentication, mood detection, music, flashcards, voice interaction, posture monitoring, and analytics are managed through separate application providers. Local persistence through Hive and Shared references allows several learning functions to remain available when network connectivity is limited. The system also incorporates BFS, DFS, and A\* algorithms to organize topic dependencies and recommend feasible learning paths. The contribution of BusMind is primarily architectural and functional. The proposed system demonstrates how multiple lightweight AI, learning, and mobile-sensing capabilities can be combined into a single commute-oriented learning workflow. It does not claim clinical, psychological, or statistically validated educational benefits without controlled user studies.

**Keywords:** Mobile learning; Emotion-aware computing; Personalized learning; Facial expression recognition; Spaced repetition; Voice summarization; Posture monitoring; Learning analytics; Flutter

## I. INTRODUCTION

Mobile devices have become an important part of the daily learning environment of university students. Their portability allows learning activities to continue outside classrooms, laboratories, and homes, making mobile learning particularly suitable for situations in which students have short periods of otherwise unused time. A systematic review by Crompton and Burke reported extensive use of mobile learning in higher education and highlighted the potential of mobile devices to extend learning beyond conventional educational settings [1]. However, the availability of a mobile device alone does not ensure that fragmented periods such as daily bus journeys become productive learning sessions.

A bus journey differs significantly from a conventional study environment. The learner may have only a few minutes available, the surrounding environment may be noisy or distracting, internet connectivity may

fluctuate, and the learner's attention and energy may vary throughout the journey. A learning system designed for such an environment therefore needs to be flexible rather than assuming that every session should follow the same learning pattern.

BusMind is proposed as a context-aware mobile study companion that attempts to address this problem. Instead of functioning as a conventional collection of learning tools, the system combines several services within a single workflow. The learner can review flashcards, listen to recommended music, capture study notes using voice, monitor posture, and inspect learning progress without moving between multiple applications.

An important characteristic of BusMind is that it treats the learner's context as an input to the system. Facial-expression analysis provides an optional contextual signal, while available time, learning history, topic priority, and user preference provide additional information. This follows the broader concept of context-aware recommendation, where recommendations are generated by considering the situation in which the user is interacting with the system [2].

The objective is not to claim that a facial expression completely describes a learner's emotional state. Instead, the detected expression is converted into a coarse application-level context that may influence recommendations. This distinction is important because emotion recognition remains sensitive to factors such as lighting, facial orientation, dataset characteristics, and individual differences.

The learning component of BusMind is based on retrieval and spacing principles. Retrieval practice requires learners to actively recall information, while spacing distributes learning opportunities over time. Research has consistently identified these strategies as useful mechanisms for supporting long-term learning and retention [3], [4]. BusMind applies these principles through an adaptive flashcard scheduler that gives greater attention to cards with lower recent recall or longer review intervals.

The major contribution of the proposed system is therefore the integration of these capabilities around a specific real-world context: learning during bus travel. Rather than introducing another isolated flashcard, recommendation, or sensing application, BusMind explores how these components can cooperate within a lightweight mobile architecture.

## **II. RELATED WORK**

### **A. Mobile Learning**

Mobile learning enables educational interaction through portable devices across different locations and contexts. Crompton and Burke's systematic review of mobile learning in higher education identified learning achievement, learner perception, pedagogy, and other dimensions as important research areas [1]. Their findings demonstrate that mobile devices can extend learning opportunities beyond formal classrooms.

However, many existing mobile-learning applications primarily focus on content access. BusMind extends this concept by considering the conditions under which learning occurs. The proposed system is designed for short, repeated, and potentially interrupted sessions,

making activity selection and persistence important components of the architecture.

### **B. Context-Aware Recommendation**

Traditional recommendation systems commonly rely on user preferences and historical interactions. Context-aware recommender systems additionally incorporate information such as time, location, activity, or environmental conditions [2]. This principle is particularly relevant to BusMind because the usefulness of an activity can depend on the remaining commute time and current learning state.

For example, a learner with ten minutes remaining may benefit from a small flashcard session, whereas a learner with a longer journey may choose voice-based note creation or a deeper study activity. The proposed recommendation model therefore combines context with explicit user preferences instead of relying solely on mood.

### **C. Emotion-Aware Music Recommendation**

Emotion and contextual information have increasingly been incorporated into music recommendation research. Assuncao, Piccolo, and Zaina reviewed research involving emotional and contextual information in music recommendation and identified user preference, activity, satisfaction, cognitive load, and other contextual variables as relevant factors [5]. BusMind adopts this general principle while keeping its recommendation mechanism lightweight.

The detected expression is not directly treated as a psychological measurement. Instead, it is converted into an application-level state that can influence candidate music selection. This reduces the risk of overinterpreting the output of the facial-expression model.

### **D. Retrieval Practice and Spaced Learning**

Retrieval practice involves recalling previously learned information rather than simply reviewing it. A systematic review of applied retrieval-practice research reported consistent benefits across different educational conditions [4]. Similarly, Carpenter, Pan, and Butler described spacing and retrieval practice as well-supported learning strategies across different contexts [3].

BusMind incorporates these principles into its flashcard system. Cards are stored with review history and can be assigned a priority according to recall performance, difficulty, and elapsed time since previous review.

## E. Vision Transformers for Facial Expression Recognition

Vision Transformers apply the Transformer architecture to sequences of image patches. Dosovitskiy et al. demonstrated that image classification can be performed by representing an image as a sequence of fixed-size patches and processing the sequence through Transformer layers [6]. BusMind uses a Vision Transformer-based facial-expression model to obtain an expression category from an optional camera frame.

The use of a Transformer does not eliminate the limitations of facial-expression recognition. Model output remains dependent on image quality, facial visibility, training data, and environmental conditions.

## F. Smartphone Inertial Sensing

Smartphones contain accelerometers and gyroscopes that provide measurements along the device axes. Reviews of smartphone-based human activity recognition show that inertial sensors are widely used for capturing movement and orientation-related information [7]. BusMind uses these sensors to provide posture-awareness feedback rather than attempting to perform clinical posture assessment.

## G. Research Gap

The literature demonstrates that mobile learning, contextual recommendation, emotion-aware interaction, retrieval practice, and smartphone sensing have each been studied independently. The gap addressed by BusMind is the integration of these capabilities into a single commute-oriented learning workflow. The proposed system focuses on architectural integration, contextual activity selection, and practical mobile usability while avoiding unsupported claims about educational or medical outcomes.

## III. PROPOSED SYSTEM

### A. System Overview

BusMind is organized as a modular mobile application consisting of a presentation layer, state-management layer, service layer, persistence layer, and analytics layer. The application is implemented using Flutter and Dart, allowing the same application architecture to support mobile deployment while maintaining a consistent user interface.

The presentation layer provides the learner with access to authentication, learning activities, recommendations, flashcards, voice interaction, posture feedback, and analytics. Provider-based state management separates screen-level presentation from application logic.

Service components handle functions such as emotion inference, music recommendation, speech recognition, and sensor processing.

The persistence layer stores structured learning information locally. Hive is used for data requiring structured storage, while SharedPreferences is suitable for lightweight preferences and configuration values. This separation allows learning activities such as flashcard review to remain accessible even when network connectivity is temporarily unavailable.

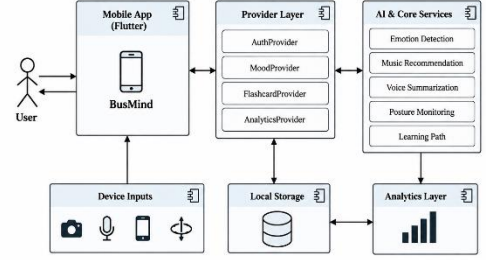


FIGURE 1. Overall architecture of the proposed BusMind mobile learning system.

### B. Learner Context Model

The system represents learner context using profile, current-session, and historical information. Let the learner state be represented as

$$X = [X_p, X_c, X_h]$$

where ( $X_p$ ) represents relatively stable profile information, ( $X_c$ ) represents current contextual information, and ( $X_h$ ) represents recent learning history.

The current context can be expressed as

$$X_c = [m, t, q, p, s]$$

where ( $m$ ) denotes the detected application-level mood, ( $t$ ) represents available session time, ( $q$ ) represents topic priority, ( $p$ ) represents user preference, and ( $s$ ) represents the current session state.

This representation prevents a single signal from controlling the complete recommendation process.

## IV. EMOTION-AWARE INTERACTION

### A. Facial Expression Detection

The emotion module optionally captures a facial image using the smartphone camera. The image is passed to

the selected facial-expression recognition model, which produces probabilities for predefined expression categories. The model used in the prototype is based on a Vision Transformer and is configured for seven expression categories.

For an input image ( $I$ ), the classifier produces

$$P(e_k | I), \quad k = 1, 2, \dots, K$$

where ( $K$ ) represents the number of expression categories. The predicted expression is

$$e^* = \arg \max_{e_k} P(e_k | I)$$

]

The predicted expression is then converted into an application-level mood state.

**Table I. Mapping of facial-expression categories to broad application-level contextual states.**

| Facial expression | Application context     |
|-------------------|-------------------------|
| Happy             | Positive / energetic    |
| Neutral           | Neutral / focused       |
| Sad               | Low-energy / supportive |
| Angry             | High-tension            |
| Fear              | Uneasy                  |
| Surprise          | Alert                   |
| Disgust           | Negative                |

The mapping is intentionally broad. It does not claim that a particular facial expression proves a user's psychological or emotional condition

This design follows the broader principle of contextual recommendation rather than making music selection depend entirely on facial-expression output.

For a candidate music group ( $g_j$ ), a conceptual recommendation score is defined as

$$R(g_j) = w_m S_m + w_p S_p + w_t S_t + w_h S_h$$

where ( $S_m$ ) represents mood compatibility, ( $S_p$ ) represents user preference, ( $S_t$ ) represents time compatibility, and ( $S_h$ ) represents historical interaction. The weights ( $w_m, w_p, w_t, w_h$ ) determine the relative contribution of each factor.

The selected candidate can then be represented as

$$g^* = \arg \max_{g_j} R(g_j)$$

This formulation also allows explicit user preferences to override an unsuitable recommendation generated from a noisy contextual signal.

## V. ADAPTIVE LEARNING MODULE

### A. Smart Flashcards

The flashcard component is designed around active recall. Each card contains learning content together with review metadata such as previous performance, difficulty, and last-review time.

The priority of a flashcard can be represented as

$$P_i = w_1(1 - A_i) + w_2 D_i + w_3 T_i + w_4 M_i$$

where ( $A_i$ ) represents recent recall accuracy, ( $D_i$ ) represents estimated difficulty, ( $T_i$ ) represents normalized elapsed time, and ( $M_i$ ) represents the learner's mastery gap.

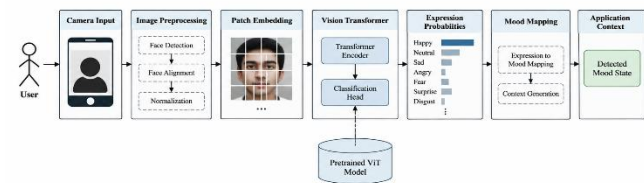
A normalized time factor can be calculated using

$$T_i' = \frac{T_i - T_{\min}}{T_{\max} - T_{\min}}$$

A card with poor recall, high difficulty, or a longer review gap therefore receives greater priority.

**Table II. Flashcard metadata used for adaptive review.**

| Parameter      | Purpose                                          |
|----------------|--------------------------------------------------|
| Difficulty     | Represents perceived or assigned card difficulty |
| Recall history | Represents previous learner performance          |
| Last review    | Determines elapsed review time                   |
| Review count   | Represents exposure frequency                    |



**Fig. 2.** Facial-expression detection and contextual mood generation pipeline.

**FIGURE 2.** Facial-expression processing and contextual mood generation pipeline.

### B. Mood-Aware Music Recommendation

The music component combines the application-level mood with user preference and contextual information.

|                  |                                    |
|------------------|------------------------------------|
| Mastery estimate | Represents current learning status |
|------------------|------------------------------------|

The flashcard state is stored locally, allowing the learner to continue reviewing cards even when network access is unavailable.

## B. Spaced Review

The review mechanism follows the principle that successfully recalled information can be revisited at increasing intervals, while incorrectly recalled information should receive earlier attention.

A simplified review interval can be expressed as

$$I_{next} = I_{current}(1 + \alpha C - \beta E)$$

where (C) represents successful recall, (E) represents an error signal, and ( $\alpha$ ) and ( $\beta$ ) are tuning parameters.

This formulation is an application-level scheduling mechanism rather than an implementation of a complete established spaced-repetition algorithm.

## C. Voice-Based Study Notes

Voice interaction is provided to reduce the amount of typing required during travel. The learner can speak a study-related note, after which speech recognition converts the audio into text.

The process can be represented as

$$A_{voice} \rightarrow T_{speech} \rightarrow T_{clean} \rightarrow S_{note}$$

where ( $A_{voice}$ ) represents the audio signal, ( $T_{speech}$ ) represents recognized speech, ( $T_{clean}$ ) represents processed text, and ( $S_{note}$ ) represents the generated study note.

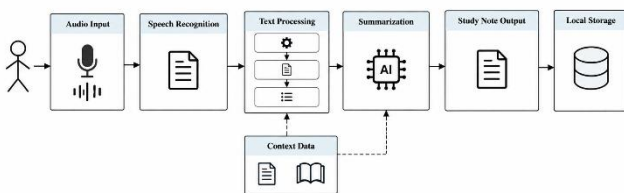


FIGURE 3. Voice-based study-note creation pipeline.

The separation of speech recognition and summarization allows either service to be improved independently.

## VI. POSTURE-AWARE MOBILE SENSING

### A. Sensor Model

The posture component uses accelerometer and gyroscope measurements. At time (t), the accelerometer vector is represented as

$$a(t) = [a_x, a_y, a_z]$$

and the gyroscope vector is represented as

$$g(t) = [g_x, g_y, g_z]$$

The combined sensor vector is

$$F(t) = [a_x, a_y, a_z, g_x, g_y, g_z]$$

The acceleration magnitude can be calculated as

$$A(t) = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

while the angular velocity magnitude can be calculated as

$$G(t) = \sqrt{g_x^2 + g_y^2 + g_z^2}$$

Rather than making decisions from individual readings, the system considers a temporal window

$$W_t = \{F(t - n + 1), \dots, F(t)\}$$

This reduces the influence of individual noisy measurements.

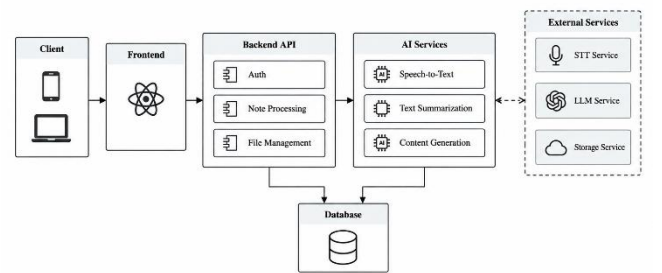


FIGURE 4. Sensor-processing architecture for posture-awareness feedback.

The current implementation is intended to provide awareness reminders and should not be interpreted as a medical or clinical posture assessment.

## VII. CONTEXT-AWARE STUDY RECOMMENDATION

The central decision-making component of BusMind is the activity recommendation model. Instead of recommending an activity only from the learner's mood, the system combines multiple contextual factors.

Let the set of candidate activities be

$$A = \{a_1, a_2, \dots, a_n\}$$

For each activity, the system calculates

$$Score(a) = w_1L(a) + w_2C(a) + w_3P(a) + w_4M(a)$$

where  $(L(a))$  represents learning value,  $(C(a))$  represents contextual suitability,  $(P(a))$  represents user preference, and  $(M(a))$  represents mood compatibility.

The selected activity is

$$a^* = \arg \max_{a \in A} Score(a)$$

This design allows the application to recommend different activities according to the learner's available time and current learning state.

For example, if the remaining journey is short, a small flashcard session may receive a higher contextual score. If more time is available, the recommendation may shift toward voice-based note generation or a longer learning activity.

**Table III. Contextual inputs used by the recommendation layer.**

| Context          | Influence on recommendation            |
|------------------|----------------------------------------|
| Available time   | Determines feasible activity duration  |
| Learning history | Identifies topics requiring review     |
| User preference  | Controls preferred activity types      |
| Mood context     | Provides an optional contextual signal |
| Topic priority   | Gives importance to pending concepts   |
| Current session  | Determines ongoing activity state      |

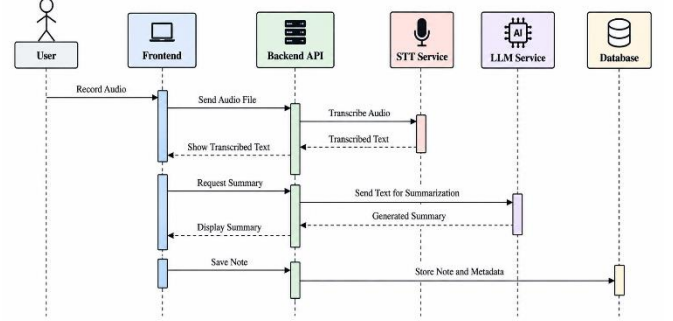
## VIII. GRAPH-BASED STUDY PATH

Learning topics are represented as a directed graph

$$G = (V, E)$$

where  $(V)$  represents topics and  $(E)$  represents prerequisite or recommended relationships.

A topic such as “Machine Learning” may depend on concepts such as probability, statistics, and linear algebra. Representing these dependencies as a graph allows BusMind to reason about the order in which topics can be studied.



**FIGURE 5. Graph-based representation of prerequisite relationships and study-path generation.**

Breadth-first search can identify shallow prerequisite relationships, while depth-first search can explore deeper dependency chains. A\* can incorporate estimated study cost when a preferred path must be selected.

The A\* evaluation function is

$$f(n) = g(n) + h(n)$$

where  $(g(n))$  represents accumulated cost and  $(h(n))$  represents estimated remaining cost.

A learning-path cost can be defined as

$$g(n) = w_1T_n + w_2D_n + w_3R_n$$

where  $(T_n)$  is estimated study time,  $(D_n)$  is topic difficulty, and  $(R_n)$  is review priority.

This allows the path generator to prefer topics that are both educationally relevant and practical within the learner's available time.

## IX. SYSTEM ARCHITECTURE AND IMPLEMENTATION

### A. Provider-Based Architecture

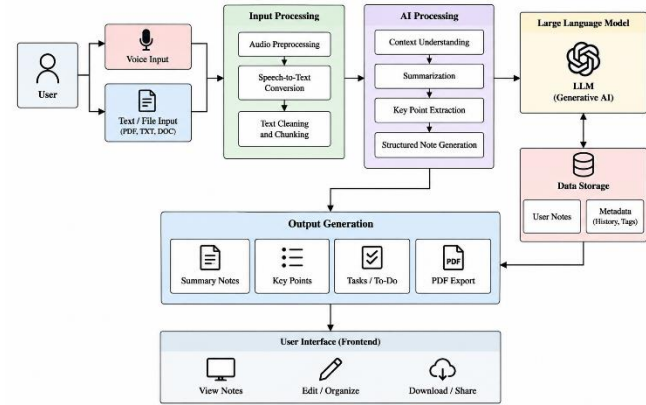
BusMind uses Provider-based state management to separate interface components from application logic. Authentication, mood detection, music, flashcards, voice interaction, posture monitoring, and analytics are represented as separate logical providers.



**Table IV. Major application providers and their responsibilities.**

| Provider              | Responsibility                   |
|-----------------------|----------------------------------|
| AuthProvider          | Authentication and session state |
| AppStateProvider      | Global application state         |
| MoodDetectionProvider | Expression and mood state        |
| MusicProvider         | Recommendation and playback      |
| FlashcardProvider     | Flashcard state and review       |
| VoiceProvider         | Speech and note generation       |
| PostureProvider       | Sensor processing                |
| AnalyticsProvider     | Learning-event aggregation       |

This architecture allows individual services to be modified without requiring substantial changes to unrelated parts of the application.



**FIGURE 6.** Provider-based software architecture of *BusMind*.

## B. Local Persistence and Offline Operation

Commuting environments can involve intermittent network connectivity. *BusMind* therefore separates network-dependent services from locally available learning functions. Flashcard information, review history, preferences, and selected analytics can remain accessible through local persistence.

Hive is used for structured application information, while lightweight configuration values can be maintained through `SharedPreferences`. The

architecture follows a graceful-degradation principle in which the failure of an optional service does not disable the entire application.

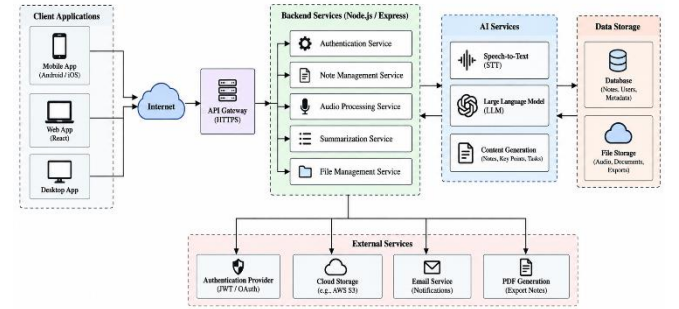
For example, if camera access is denied, the learner can continue using flashcards and voice interaction. If an external AI service is temporarily unavailable, locally stored learning content can still be accessed.

## C. Data Flow

The overall data flow can be represented as

$$\begin{aligned} \text{Input} &\rightarrow \text{Processing} \rightarrow \text{Context} \\ &\rightarrow \text{Recommendation} \rightarrow \text{Activity} \\ &\rightarrow \text{Feedback} \end{aligned}$$

The feedback generated from the activity is stored as part of the learner's history. This history then becomes an input for subsequent recommendations.



**FIGURE 7.** End-to-end data flow and feedback loop in *BusMind*.

## X. SECURITY, PRIVACY, AND RESPONSIBLE AI

*BusMind* processes several types of potentially sensitive information, including facial images, voice input, learning history, and sensor information. The system should therefore follow data-minimization principles and collect only information required for the selected functionality.

Camera and microphone features should remain permission-controlled. Raw images and audio should not be retained unnecessarily when derived information is sufficient for the application. Network communication should use secure transport, while authentication information should be protected using appropriate platform mechanisms.

Emotion recognition requires particular caution. A facial expression should not be presented as proof of a psychological state, mental-health condition, or personality characteristic. *BusMind* therefore uses expression recognition only as a contextual input to recommendation.

Similarly, posture feedback is intended to increase awareness of device orientation and prolonged patterns. It should not be represented as medical diagnosis, rehabilitation guidance, or clinical assessment.

## XI. EXPERIMENTAL EVALUATION

The current prototype is evaluated primarily at the functional level because the main objective is to demonstrate integration of the proposed modules. Each component can be verified by examining whether its expected input is correctly transformed into the required output.

**Table V. Functional verification of the major BusMind modules.**

| Module               | Input                        | Expected output                |
|----------------------|------------------------------|--------------------------------|
| Authentication       | User credentials             | Valid application session      |
| Emotion detection    | Camera image                 | Expression and contextual mood |
| Music recommendation | Mood and preferences         | Candidate recommendation       |
| Flashcards           | Question and review response | Updated review state           |
| Voice interaction    | Spoken input                 | Recognized text                |
| Posture monitoring   | Sensor sequence              | Awareness feedback             |
| Analytics            | Activity event               | Updated progress information   |
| Study path           | Topic graph                  | Recommended topic sequence     |

The evaluation can be extended into quantitative testing once sufficient user interaction data is collected. For flashcards, recall accuracy can be represented as

$$Accuracy_{recall} = \frac{Correct\ Responses}{Total\ Responses} \times 100$$

The activity completion rate can be expressed as

$$Completion = \frac{Completed\ Activities}{Started\ Activities} \times 100$$

For delayed evaluation, retention can be measured as

$$Retention = \frac{Delayed\ Correct\ Responses}{Delayed\ Questions} \times 100$$

For emotion classification, standard multi-class metrics such as precision, recall, and F1-score can be used.

$$Precision = \frac{TP}{TP + FP}$$

$$F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$$

These measures should be calculated using an appropriate labeled dataset and should not be replaced with fabricated prototype accuracy values.

## XII. RESULTS AND DISCUSSION

The implemented prototype demonstrates the feasibility of combining learning, recommendation, sensing, and AI services within a single mobile application. The modular design allows individual components to operate without tightly coupling the complete system to one service.

The emotion module provides a contextual signal that can influence recommendations, while the music module combines that signal with user preferences. The flashcard module supports retrieval-based learning and records review information locally. Voice interaction reduces the need for typing during travel, while sensor processing provides posture-awareness feedback.

A major advantage of the architecture is its ability to handle different levels of connectivity. Core learning functions do not need to depend continuously on external services. This is particularly important for a bus-oriented application because network quality can vary during travel.

The recommendation architecture also avoids making a single prediction responsible for the learner's complete experience. Mood, learning history, time availability, and preference are considered together. This makes the system more robust conceptually and provides a clearer path toward future personalization.

However, the current implementation should not be interpreted as evidence that BusMind improves academic performance or wellbeing. Such claims require controlled experiments involving sufficient participants, standardized learning material, appropriate comparison groups, and statistically meaningful analysis.



### **XIII. LIMITATIONS**

The primary limitation of the current system is that several components rely on heuristic or contextual decisions rather than extensively trained personalized models. Facial-expression recognition can be affected by lighting, camera position, facial orientation, and differences between training data and real-world users. Consequently, the detected expression should be treated as an uncertain contextual signal.

The recommendation weights also require empirical calibration. The current formulation provides a framework for combining contextual information but does not claim that the selected weights are universally optimal.

Voice interaction can introduce speech-recognition errors, particularly in noisy bus environments. Summarization quality can also vary according to the clarity and technical complexity of the spoken content.

Posture monitoring is similarly affected by smartphone position. A phone held in different orientations can produce substantially different sensor measurements even when the learner's physical posture remains similar. The current module therefore provides awareness-oriented feedback rather than medical assessment.

Finally, the current prototype does not include a controlled longitudinal study. The educational effectiveness, sustained engagement, and personalization quality of BusMind therefore remain subjects for future empirical evaluation.

### **XIV. FUTURE WORK**

Future development will focus on strengthening the personalization layer through real interaction data. Instead of using manually selected recommendation weights, future versions can learn preference patterns from user feedback while maintaining explicit controls that allow learners to override recommendations.

The emotion component can be improved by evaluating different facial-expression models under realistic mobile conditions. Privacy-preserving approaches could also be investigated so that raw facial images do not need to leave the device.

The learning engine can be extended by incorporating stronger spaced-repetition models and learner mastery estimation. The graph-based learning component can also be connected with mastery information so that prerequisite paths are generated according to actual learner weaknesses.

Posture monitoring can be improved by collecting annotated sensor sequences and training a lightweight classification model. This could reduce dependence on manually selected thresholds while preserving the application's low-resource design.

A future controlled study should compare BusMind with a conventional mobile-learning application. The study can measure recall, delayed retention, activity completion, session consistency, perceived usefulness, and user satisfaction. Such an evaluation would provide evidence for determining whether contextual personalization produces measurable benefits.

The system can also be extended to other fragmented learning environments, such as metro journeys, waiting periods, campus travel, and short breaks between academic activities.

### **XV. CONCLUSION**

BusMind presents a context-aware mobile learning architecture designed specifically for students who spend significant amounts of time travelling by bus. The proposed system combines facial-expression-based contextual sensing, mood-aware music recommendation, adaptive flashcards, voice-based study notes, posture awareness, analytics, and graph-based study-path generation within a unified Flutter application.

The system is designed around the observation that commuting does not provide the same learning conditions as a conventional study environment. Available time, attention, connectivity, and user state can change continuously. BusMind therefore treats these factors as contextual information that can influence activity selection rather than assuming a fixed learning sequence.

The integration of retrieval practice and spaced review provides an established learning foundation for the flashcard component, while contextual recommendation allows the application to adapt its interactions according to the learner's current situation. Smartphone sensors and optional facial-expression analysis further expand the range of contextual information available to the system.

The architecture also emphasizes modularity, local persistence, graceful degradation, and responsible use of AI. In particular, the system avoids treating facial expressions as psychological diagnoses or posture feedback as medical assessment.

The current work should be considered a functional prototype and architectural contribution. Its effectiveness as a long-term educational system remains

to be established through controlled experimentation and real-world user studies. Nevertheless, the proposed architecture provides a practical foundation for combining AI-assisted personalization, mobile learning, and contextual sensing within fragmented daily travel time.

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